



LarvaLens

App Flow Navigation and User Journeys

Every route, navigation rule, entry point, core journey, empty/error state, reconnect behavior and redirect rule for the LarvaLens PWA.

Prepared for	Mohit S. and the LarvaLens team
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Vibe Document 03 - App Flow and Navigation Map

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8.1 Routes

Route	Access	Purpose
/	Public	Product explanation, limitations and sign-in CTA.
/login	Public only	Email/password sign-in and account creation.
/scan	Authenticated	Record/select video, validate, request location and submit.
/scans/:scanId/processing	Owner/reviewer	Actual stage and progress from Realtime.
/scans/:scanId	Owner/reviewer	Summary, evidence tracks, quality, model version and review state.
/history	Authenticated	Paginated own scans.
/map	Public aggregated or authenticated	Hotspot map with privacy-preserving markers.
/review	Reviewer/admin	Review queue.
/review/:scanId	Reviewer/admin	Evidence and human decision form.
/admin/models	Admin	Read-only model readiness, hashes and stage metadata.
/settings	Authenticated	Profile and privacy information.
*	Any	Helpful not-found page.

8.2 Navigation

Desktop uses a top bar with Scan, History and Map; reviewer/admin items appear only when authorized. Mobile uses a bottom navigation with Scan, History and Map. Account/settings lives in the top-right menu. The primary scan CTA remains reachable with one tap.

8.3 First-time journey

text

```
Landing
-> Create account / Sign in
-> Email/password authentication
-> /scan
-> permission explainer
-> select/record short video
-> optional location consent
-> client validation
-> POST real scan
-> /scans/{id}/processing
-> completed / failed / retake-required
-> /scans/{id}
```

8.4 Core journey 1 - successful scan

1. User selects a real 5-10 second video.
2. Web validates extension, size and available metadata.

3. User optionally grants location; denial does not block analysis.
4. API validates JWT, model readiness and media.
5. API creates `scans` row and private storage object; returns 202.
6. Web navigates using returned UUID and subscribes to that row.
7. Processing states update from actual backend writes.
8. On `completed`, web refetches result and track evidence.
9. User sees probable count, rejected count, confidence, quality, risk, limitations and map location if provided.

8.5 Core journey 2 - model unavailable

text

```
Submit
-> API checks manifest/files
-> missing/hash mismatch
-> HTTP 503 MODEL_NOT_READY
-> keep selected video locally in the page
-> show "Analysis service is not ready"
-> offer Retry and System Status
```

No scan is presented as successful. No placeholder result appears.

8.6 Core journey 3 - reviewer decision

1. Reviewer opens `/review` and sees authorized real scans needing review.
2. Reviewer opens evidence, including accepted/rejected tracks and model versions.
3. Reviewer selects confirm, reject or inconclusive and adds notes.
4. API writes a separate `reviews` record and updates `review_status` transactionally.
5. Raw counts and track probabilities remain unchanged.

8.7 Empty, error and reconnect states

Condition	UI behavior
No history	Explain how to create the first scan; no fake cards.
No map points	Show an empty map message and privacy note.
File too large/long	Explain configured limit before upload.
Unsupported/corrupt video	Ask for MP4/WebM/MOV retake or conversion.
Location denied	Continue without coordinates.
Low light/blur/camera shake	Show <code>retake_required</code> with backend reason.
Network interruption before upload	Preserve selection in page and allow retry.
Realtime disconnect	Show reconnecting state and poll/refetch current scan.
Inference failure	Show request ID and retryability; never show partial count as final.
Unauthorized route	Redirect to login or a dedicated forbidden page, preserving intended route.

8.8 Redirect rules

- Logged-in user visiting `/login` -> `/scan`.
- Logged-out user visiting protected route -> `/login?next=<encoded-route>`.
- Successful submission -> `/scans/{scanId}/processing`.
- Processing completed -> replace route with `/scans/{scanId}`.

- Logout -> / and clear query cache/subscriptions.
- Reviewer without reviewer/admin role -> /forbidden.